ADDIS ABABA SCIENCE AND TECHNOLOGY UNIVERSITY

College Of Engineering
Department Of Software Engineering
Internet Programming I
Final Project - DineOnline
SECTION-B (Group 5)

Firaol Gula	**0588/16**
Fikremariam Debela	**0576/16**
Firaol Beshada	**0587/16**
Kaleb dereje	**0769/16**
Fitsum mulugeta	**0601/16**
Kidus Bafena	**0816/16**

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Introduction

Technology is changing the way businesses operate, especially in the food and restaurant industry. Many restaurants are moving from manual systems to digital platforms in order to improve efficiency, accuracy, and customer satisfaction. Online ordering systems help restaurants manage orders, inventory, payments, and delivery in a faster and more organized way.

DineOnline is a virtual restaurant management and ordering system developed to support DineHub Ethiopia, a small but growing virtual kitchen. The system is designed to replace the current paper-based and semi-manual processes with a modern web-based solution. By using DineOnline, the restaurant can handle customer orders more effectively, reduce errors, and make better business decisions using real-time data.

Context

Background Information

DineHub Ethiopia is a privately owned virtual restaurant located in Addis Ababa. It was established in 2022 and operates without a physical dining space. Instead, it prepares meals in small kitchen facilities and sells food through social media platforms such as Instagram, Telegram, and phone calls. Orders are delivered to customers using third-party riders.

Currently, the restaurant uses a mostly manual system. Customer orders are received through direct messages, written in a paper ledger, and later typed into Microsoft Excel at the end of the day. Payment confirmation is done by checking screenshots sent by customers. Inventory levels are tracked manually, and kitchen staff depend on handwritten notes to know which orders to prepare.

As the number of customers increases, this manual approach causes many problems such as order mistakes, delays, lost records, and difficulty in tracking stock and sales. Finding old orders or generating reports also takes a long time.

Environment Where the System Will Be Used

"What is the current situation or problem in the environment where the system will be used, and how will the system fit into and improve that environment?"

The DineOnline system will be used in a real working environment of a small virtual restaurant in Ethiopia. Customers will use mobile phones, tablets, or computers to view the menu and place orders, while order desk agents and kitchen supervisors will use desktop computers or tablets inside the kitchen to manage orders and food preparation. Managers will use the system to check sales, inventory, and performance reports, and riders will use their smartphones to receive delivery details and update order status. Since internet connectivity may sometimes be unstable and most users are not highly technical, the system is designed to be simple, fast, and easy to use with little training. Overall, DineOnline .supports daily restaurant operations by improving accuracy, speed, and service quality for both staff and customers.

Motivation of the Project

The motivation for choosing the DineOnline project comes from the real problems observed in the daily operations of DineHub Ethiopia, where orders are managed using paper records and manual data entry. This manual system causes frequent errors, delays in order processing, poor inventory control, and late reporting, especially during busy hours. As customer demand continues to grow, the existing system is no longer efficient or reliable. The DineOnline project aims to solve these problems by providing a digital platform that automates ordering, payment verification, inventory tracking, and reporting, making restaurant operations faster, more accurate, and easier to manage.

Objective of the Project

"What are the general and specific objectives of this system?"

General Objective

The general objective of the DineOnline project is to develop a web-based virtual restaurant management system that replaces the existing manual processes with a digital solution, in order to improve order handling, inventory management, reporting, and overall service efficiency at DineHub Ethiopia.

Specific Objectives

The specific objectives of the project are:

- To allow customers to browse a digital menu and place orders easily using mobile phones or computers.
- To automate order management and reduce errors caused by handwritten records and manual data entry.
- To track inventory in real time and reduce stock shortages and wastage.
- To support order status tracking and delivery coordination for staff and riders.
- To generate instant sales and performance reports for managers to support better decision-making.
- To improve customer satisfaction by providing faster service and clear order tracking.

Corresponding Requirements

The system requirements of DineOnline are designed to directly support and achieve the stated objectives of the project. To meet the objective of allowing customers to place orders easily, the system must provide customer registration, login, digital menu browsing, cart management, and online order placement features.

- To reduce errors and improve order handling, the system must record all orders digitally, store them in a central database, and allow staff to view and update order status in real time. This removes the need for handwritten records and repeated data entry.

- To support effective inventory management, the system must automatically update stock levels when an order is confirmed and generate alerts when inventory falls below a defined threshold. This helps prevent stock shortages and canceled orders.
- To improve delivery coordination, the system must allow staff to assign riders, update delivery status, and enable riders to access order details using their smartphones. Customers must also be able to track their order progress.

Finally, to support management decision-making, the system must generate real-time sales, inventory, and performance reports that managers can access through a dashboard. These requirements ensure that each project objective is fully supported by the system's functionality.

Functional Requirements

The functional requirements define the specific behaviors, features, and operations that the DineOnline system must provide to meet the needs of its users. These requirements describe how the system will interact with customers, restaurant staff, and delivery personnel to ensure efficient and accurate order processing. They focus on the tasks the system must perform, such as account management, menu browsing, order placement, payment verification, inventory management, and delivery tracking. By clearly specifying these functionalities, the system can be developed to provide a seamless, secure, and user-friendly experience for all stakeholders involved in the online food ordering process.

1. The system shall allow customers to register and log in using a phone number and password.
2. The system shall allow customers to browse the digital menu and view food items, prices, and descriptions.
3. The system shall allow customers to add food items to a cart and modify quantities before placing an order.
4. The system shall allow customers to place an order and receive an order confirmation.
5. The system shall allow customers to upload payment confirmation screenshots for their orders.
6. The system shall allow order desk agents to view new orders and verify customer payment proofs.
7. The system shall allow kitchen supervisors to view a real-time list of confirmed orders and update their preparation status.
8. The system shall automatically update inventory levels when an order is confirmed.
9. The system shall generate low-stock alerts when inventory reaches a minimum level.
10. The system shall allow managers to view sales reports, inventory reports, and order summaries.
11. The system shall allow riders to view assigned deliveries and update delivery status.
12. The system shall allow customers to track their order status from preparation to delivery.

Significance of the System

The DineOnline system is important because it helps solve many problems caused by manual order handling in the restaurant. By digitizing the ordering and management process, the system reduces errors, saves time, and improves accuracy in daily operations. Staff no longer need to write orders on paper or re-enter data into Excel, which minimizes mistakes and delays.

The system also improves inventory management by updating stock levels automatically and alerting staff when items are running low. This helps prevent food shortages and customer dissatisfaction. For management, DineOnline provides instant access to sales and performance reports, making it easier to monitor business activities and make better decisions.

For customers, the system offers a faster and more transparent ordering experience, with clear order tracking and status updates. Overall, DineOnline increases efficiency, improves service quality, and supports the growth and sustainability of the restaurant.

Beneficiaries of the System

The DineOnline system benefits several users involved in the restaurant's daily operations. Customers benefit by being able to browse the menu easily, place orders online, track their order status, and receive faster and more reliable service.

- Order desk agents benefit from the system by receiving orders digitally, which reduces paperwork, minimizes errors, and makes order verification easier. Kitchen supervisors benefit from having a clear, real-time view of pending orders and inventory levels, helping them manage food preparation more efficiently.
- Managers benefit by gaining access to instant sales, inventory, and performance reports, which support better planning and decision-making. Riders also benefit from the system by receiving clear delivery instructions and updating delivery status through their smartphones. Overall, the system supports all users by improving coordination, efficiency, and service quality.

Feasibility Analysis

Technical Feasibility

Does the team have the necessary technical resources and expertise to successfully build and maintain the system?

Technical feasibility evaluates whether the project's requirements can be met with the current hardware, software, and technical skills available to the team.

- **Hardware and Software:** The system can run on any PC with at least 4 GB of RAM and utilizes a modern software stack including Node.js, Express, React, and MongoDB Atlas.

- **Development Tools:** The team uses industry-standard tools such as VS Code, GitHub, Figma, and Postman.
- **Team Competency:** The project is being developed by 3rd-year Software Engineering students who are already skilled in the MERN stack.
- **Conclusion:** The project is technically feasible because all necessary resources are available with zero license costs.

2. Economic Feasibility

Is the project financially viable, and does the expected return on investment justify the costs?

Economic feasibility is a cost-benefit analysis that determines if the financial gains of the system outweigh the expenses required for development and operation

- **Development Costs:** Because the team is self-developing the software, there are no external developer stipends or laptop rental costs, resulting in a total initial outlay of 0 ETB.
- **Annual Benefits:** The system is projected to save approximately 125,620 ETB per year by reducing manual labor (16,420 ETB), minimizing order refunds (31,200 ETB), and increasing customer retention (78,000 ETB).
- **Conclusion:** The project is extremely cost-effective with an "immediate" payback period, making it highly feasible economically.

3. Operational Feasibility

How well will the system integrate into the daily business workflow and be accepted by the people who use it?

Operational feasibility assesses how well the proposed solution fits the current business environment and whether the staff will support and use the new system.

- **Stakeholder Acceptance:** Staff have expressed enthusiasm during initial demonstrations, and management has pledged full cooperation because the system reduces workload without eliminating jobs.
- **Change Management:** To ensure a smooth transition, the project includes a plan with two-hour onboarding sessions and quick reference guides for all users.
- **Conclusion:** The system is considered operationally viable and well-accepted by its intended users.

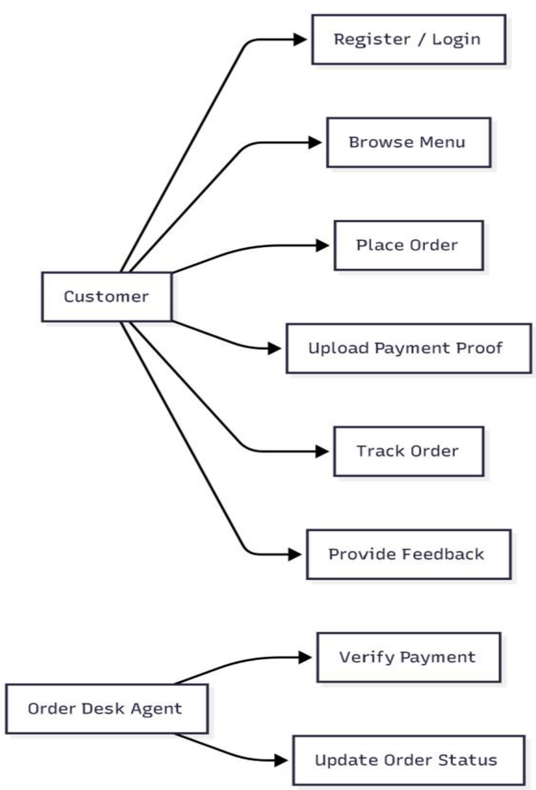
Software Requirements Specification (SRS)

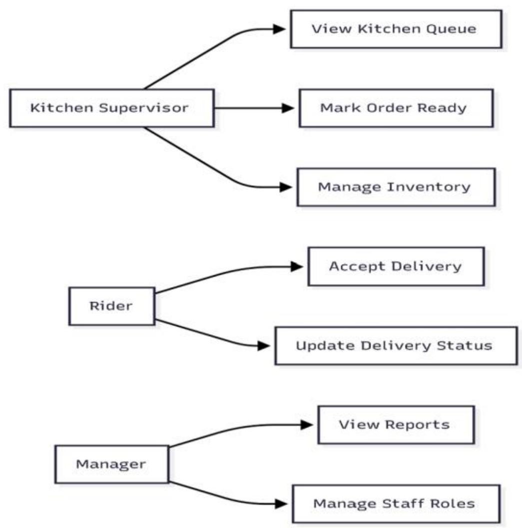
An SRS is the foundational document of system design that outlines both the functional and non-functional needs of an organization. It acts as a contract between the stakeholders (like the restaurant manager) and the developers (the software engineering students) to ensure everyone agrees on the system's features before any code is written. In this project, the SRS:

Use Case Diagram

- The Use Case Diagram illustrates the interactions between system actors (Customer, Order Desk Agent, Kitchen Supervisor, Rider, and Manager) and the Dine-online system. It shows what actions each user can perform and defines the system's functional scope.

Customer: Registers/Logs in, browses the interactive menu, manages the cart, uploads payment proof, tracks orders, and provides feedback.
Order Desk Agent: Accesses the incoming order list, verifies payment screenshots, and updates order status.
Kitchen Supervisor: Views the real-time kitchen queue, marks orders as "Ready," and monitors low-stock inventory alerts.
Rider: Receives delivery assignments and provides live GPS/progress updates to the customer.
General Manager: Accesses the performance dashboard, reviews daily/weekly sales reports, and manages staff roles.
This diagram helps identify system boundaries and ensures that all user requirements are addressed.





Problems and limitations of the existing system

What are the specific weaknesses of the manual process that hinder the growth of DineHub Ethiopia?

The current manual setup, while functional for a small startup, has reached its limit. The reliance on paper ledgers and social media DMs creates a "bottleneck" that prevents the business from scaling efficiently. Below are the primary problems and limitations:

- **The "Double-Work" Burden (Data Redundancy):** Staff currently face the exhausting task of writing orders by hand in a ledger and then re-typing those same details into Excel at the end of the night. This wastes approximately **45 minutes of productive time every day** that could be spent on customer service or kitchen prep.
- **The Cost of Human Error:** Handwritten notes are often messy or misinterpreted during the rush of lunch hour. These transcription errors lead to an average of **three incorrect meals per week**, resulting in wasted ingredients, lost revenue through refunds, and dissatisfied customers.
- **"Blind" Inventory Management:** Because there is no live link between the kitchen and the ordering desk, agents often sell meals that are already out of stock. Roughly **five customers every week** pay for items they cannot receive, forcing the staff to make awkward "apology calls" to change or cancel orders.
- **The Information Lag:** Management is essentially "flying blind" during the business day. Because sales reports are only compiled manually 24 hours late, the owner cannot make quick decisions regarding daily promotions, staffing needs, or emergency ingredient restocking.

- **Security & Physical Vulnerability:** A single spilled cup of coffee or a misplaced notebook could erase an entire month of financial records. Physical ledgers offer **zero data security**, making it impossible to track who accessed sensitive customer information or to recover data in the event of a physical accident.
- **Inefficient Retrieval:** Searching for a specific past order or a customer's history requires flipping through stacks of paper. What should take a two-second digital search currently takes **10 minutes or more**, slowing down dispute resolutions and personalized marketing efforts.

The Proposed System: DineOnline

The proposed system, **DineOnline**, is a comprehensive web-based platform built using the **MERN stack** (MongoDB, Express, React, and Node.js). It is designed to replace the existing manual, paper-led workflow with a fully integrated digital solution that connects customers, kitchen staff, delivery riders, and management in real-time.

How it improves the Existing System

What specific operational benefits does DineOnline provide over the manual process?

DineOnline is designed as a **Business Process Improvement (BPI)** solution, meaning it optimizes the existing workflow rather than radically changing the business structure. Key improvements include:

- **Elimination of Manual Redundancy:** By capturing orders digitally at the source (the customer), the system eliminates the need for staff to handwrite orders and then re-type them into Excel—a process that currently wastes **45 minutes every day**.
- **Drastic Error Reduction:** Automated digital logging is projected to reduce manual transcription errors by **90%**, significantly cutting down on the three incorrect meals and subsequent refunds typically processed each week.
- **Proactive Inventory Control:** Unlike the manual system which has no real-time tracking, DineOnline automatically deducts ingredients as orders are confirmed. This is expected to result in **30% fewer stock-outs**, preventing situations where customers pay for items that are unavailable.
- **Instant Business Intelligence:** Management no longer has to wait for a 24-hour lag to see sales data. The new system provides **instant reporting**, allowing the manager to see top-selling meals and customer trends as they happen.
- **Enhanced Customer Transparency:** Customers gain peace of mind through **live order tracking** and automated notifications, reducing the need for repeated phone calls to check on a meal's status.

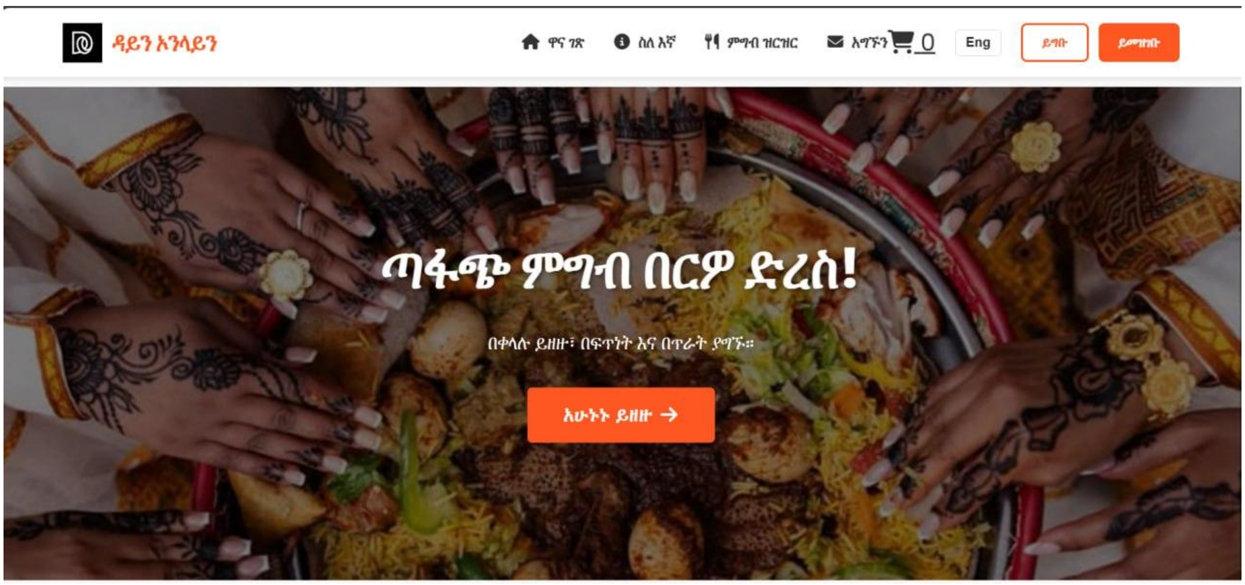
- **Data Security and Integrity:** The platform replaces vulnerable paper ledgers with a secure cloud database. It uses **Role-Based Access Control (RBAC)** to ensure that sensitive information—like financial reports or customer data—is only accessible to authorized personnel

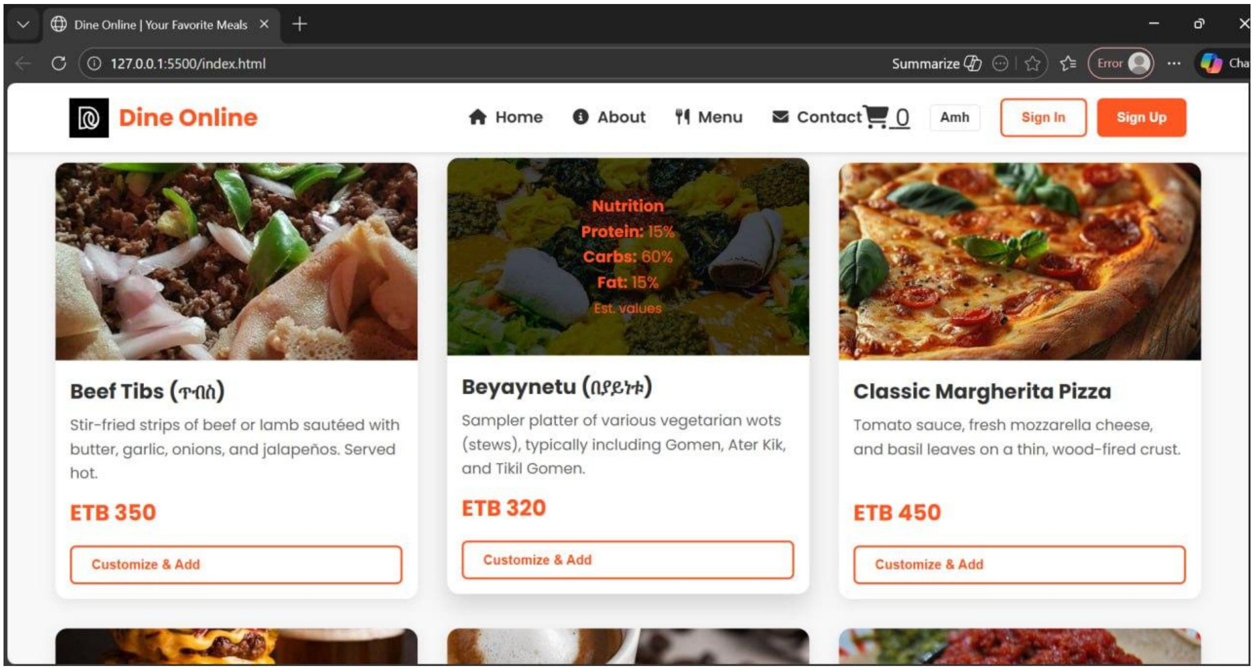
Adaptive Featured Products

How does the system personalize the menu for users to increase sales and satisfaction?

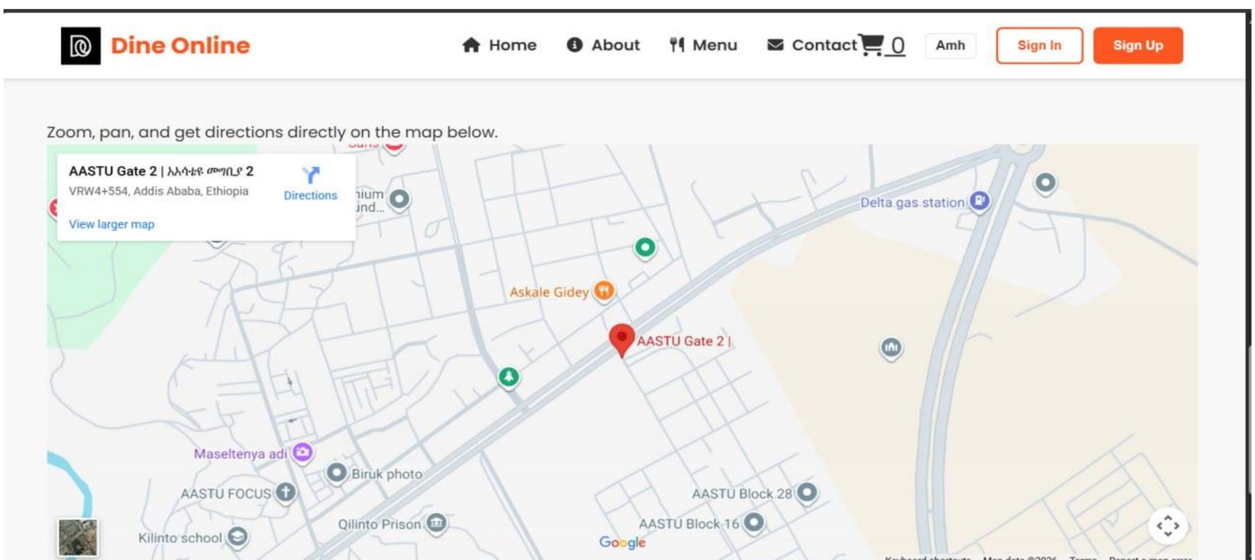
DineOnline is designed to be more than a static ordering page; it acts as an intelligent sales assistant. By analyzing past order history and real-time inventory levels, the system dynamically shifts which items appear in the "Featured" or "Recommended" sections of the dashboard. This ensures that the first things a customer sees are the items they are most likely to buy, creating a personalized shopping experience that mimics the attentiveness of a physical restaurant server.

Demonstration





Dine Online | Your Favorite Meals
127.0.0.1:5500/index.html
Dine Online
Home
About
Menu
Contact
0
Amh
Sign In
Sign Up
Nutrition
Protein: 15%
Carbs: 60%
Fat: 15%
Est. values
Beef Tibs (ጥብስ)
Stir-fried strips of beef or lamb sautéed with butter, garlic, onions, and jalapeños. Served hot.
ETB 350
Customize & Add
Beyaynetu (በያይነቱ)
Sampler platter of various vegetarian wots (stews), typically including Gomen, Ater Kik, and Tikil Gomen.
ETB 320
Customize & Add
Classic Margherita Pizza
Tomato sauce, fresh mozzarella cheese, and basil leaves on a thin, wood-fired crust.
ETB 450
Customize & Add



Dine Online
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